

SLIDELL, LA, USA

WMO: 722330

Lat: 30.343N

Lon: 89.822W

Elev: 27

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53865

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                       |           |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>27.8 | (c)<br>30.4 | (d)<br>15.5                   | (e)<br>12.1 | (f)<br>38.3 | (g)<br>19.5 | (h)<br>14.7 | (i)<br>38.6 | (j)<br>19.0           | (k)<br>52.4 | (l)<br>17.5 | (m)<br>54.7 | (n)<br>2.7            | (o)<br>10 | (p)<br>0.360 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD<br>to 0.4% DB |            |
|---------------|---------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|-------------------------|------------|
|               |                           | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                         |            |
|               |                           | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                    | PCWD       |
| (a)<br>8      | (b)<br>16.7               | (c)<br>93.9     | (d)<br>77.4 | (e)<br>92.4 | (f)<br>77.3 | (g)<br>90.9 | (h)<br>77.0 | (i)<br>80.6         | (j)<br>88.3 | (k)<br>79.9 | (l)<br>87.6 | (m)<br>79.1 | (n)<br>86.8 | (o)<br>6.2              | (p)<br>230 |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 79.0                            | 150.5 | 84.0 | 77.8 | 144.7 | 83.6 | 77.0 | 140.7 | 83.2 | 44.1          | 88.2 | 43.2 | 86.9 | 42.5 | 87.0 | 84.9              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |
| (a)               | (b)  | (c)  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   |
| 17.3              | 15.5 | 12.9 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
|                   |      |      | 22.1                       | 97.7 | 3.0                | 2.8 | 19.9                                               | 99.7 | 18.1       | 101.4 | 16.5       | 102.9 | 14.3       | 105.0 |
|                   |      |      | 20.4                       | 82.1 | 3.2                | 1.2 | 18.0                                               | 82.9 | 16.1       | 83.6  | 14.3       | 84.3  | 12.0       | 85.2  |

Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 68.0   | 51.4  | 55.5  | 61.5  | 67.5  | 74.5  | 80.4  | 81.8  | 82.1  | 78.3  | 69.1  | 59.0  | 53.8 |
|                                                                                  | DBStd       | 13.07  | 10.08 | 9.46  | 8.36  | 6.53  | 5.35  | 2.99  | 2.36  | 2.93  | 4.07  | 7.63  | 8.48  | 9.70 |
|                                                                                  | HDD50       | 255    | 107   | 46    | 12    | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 18    | 71   |
|                                                                                  | HDD65       | 1579   | 434   | 289   | 167   | 48    | 6     | 0     | 0     | 0     | 0     | 50    | 216   | 369  |
|                                                                                  | CDD50       | 6816   | 150   | 200   | 368   | 527   | 759   | 913   | 987   | 995   | 850   | 591   | 288   | 188  |
|                                                                                  | CDD65       | 2665   | 12    | 23    | 57    | 124   | 300   | 463   | 522   | 530   | 401   | 175   | 37    | 21   |
|                                                                                  | CDH74       | 25459  | 18    | 78    | 288   | 961   | 2933  | 4866  | 5427  | 5491  | 3630  | 1494  | 222   | 51   |
|                                                                                  | CDH80       | 9938   | 0     | 2     | 19    | 139   | 955   | 2147  | 2392  | 2433  | 1435  | 399   | 17    | 0    |
| Wind                                                                             | WSAvg       | 5.0    | 5.8   | 6.2   | 6.1   | 6.2   | 5.4   | 4.2   | 3.4   | 3.4   | 4.0   | 4.5   | 4.9   | 5.6  |
| Precipitation                                                                    | PrecAvg     | 63.3   | 5.9   | 4.3   | 5.3   | 4.7   | 5.2   | 6.1   | 7.2   | 6.1   | 4.7   | 4.0   | 4.5   | 5.1  |
|                                                                                  | PrecMax     | 93.0   | 18.1  | 8.7   | 10.9  | 12.4  | 21.3  | 20.6  | 13.9  | 10.8  | 13.4  | 12.0  | 13.0  | 18.3 |
|                                                                                  | PrecMin     | 40.0   | 0.3   | 0.8   | 0.3   | 0.1   | 0.2   | 0.6   | 2.8   | 2.7   | 1.3   | 0.0   | 0.9   | 1.9  |
|                                                                                  | PrecStd     | 13.1   | 4.4   | 2.4   | 2.9   | 2.9   | 4.9   | 4.4   | 2.8   | 2.3   | 3.3   | 3.1   | 3.4   | 3.5  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 76.3  | 80.0  | 82.0  | 85.6  | 91.2  | 95.9  | 95.5  | 96.9  | 93.9  | 90.0  | 82.1  | 78.1 |
|                                                                                  |             | MCWB   | 68.4  | 69.0  | 69.0  | 70.9  | 73.9  | 75.8  | 77.7  | 78.0  | 77.0  | 75.1  | 71.1  | 70.8 |
|                                                                                  | 2%          | DB     | 73.1  | 76.4  | 80.1  | 83.3  | 89.0  | 93.0  | 93.3  | 93.6  | 91.0  | 86.9  | 79.3  | 75.2 |
|                                                                                  |             | MCWB   | 66.7  | 67.6  | 68.1  | 71.1  | 73.9  | 76.1  | 77.8  | 78.3  | 76.2  | 73.7  | 70.2  | 69.3 |
|                                                                                  | 5%          | DB     | 70.7  | 73.2  | 77.4  | 81.5  | 87.6  | 91.0  | 91.5  | 92.1  | 89.6  | 84.2  | 76.7  | 72.5 |
|                                                                                  |             | MCWB   | 65.5  | 66.6  | 67.4  | 69.9  | 73.5  | 76.1  | 77.7  | 78.1  | 75.9  | 72.3  | 67.8  | 67.8 |
|                                                                                  | 10%         | DB     | 67.8  | 70.7  | 74.7  | 79.5  | 85.3  | 89.6  | 90.2  | 90.5  | 87.8  | 81.8  | 73.6  | 69.9 |
|                                                                                  |             | MCWB   | 63.9  | 65.3  | 66.3  | 68.5  | 72.7  | 76.0  | 77.4  | 77.9  | 75.5  | 71.2  | 66.4  | 66.3 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 71.0  | 72.0  | 72.7  | 75.6  | 77.9  | 80.8  | 81.4  | 82.3  | 79.9  | 78.5  | 75.0  | 74.0 |
|                                                                                  |             | MCDB   | 73.7  | 76.3  | 77.4  | 80.1  | 84.8  | 88.9  | 89.5  | 90.3  | 86.5  | 83.8  | 78.4  | 75.7 |
|                                                                                  | 2%          | WB     | 69.0  | 70.2  | 71.2  | 74.3  | 76.7  | 79.6  | 80.3  | 81.1  | 78.8  | 77.0  | 72.6  | 70.9 |
|                                                                                  |             | MCDB   | 71.6  | 74.0  | 75.8  | 79.2  | 83.9  | 87.7  | 88.3  | 88.7  | 85.5  | 82.8  | 76.6  | 73.6 |
|                                                                                  | 5%          | WB     | 67.0  | 68.5  | 70.1  | 73.2  | 75.9  | 78.5  | 79.5  | 80.2  | 78.1  | 75.7  | 70.4  | 69.1 |
|                                                                                  |             | MCDB   | 69.6  | 72.2  | 74.8  | 78.3  | 82.9  | 86.5  | 87.4  | 87.8  | 85.1  | 81.7  | 75.0  | 71.6 |
|                                                                                  | 10%         | WB     | 64.3  | 66.2  | 68.7  | 72.0  | 75.1  | 77.8  | 78.6  | 79.3  | 77.3  | 74.2  | 68.5  | 66.7 |
|                                                                                  |             | MCDB   | 67.0  | 69.7  | 73.3  | 77.0  | 82.0  | 85.6  | 86.4  | 87.0  | 84.6  | 79.6  | 73.1  | 69.3 |
| Mean Daily<br>Temperature<br>Range                                               |             | MDBR   | 20.6  | 20.5  | 21.2  | 20.7  | 19.8  | 17.7  | 16.5  | 16.7  | 18.4  | 22.0  | 23.6  | 20.4 |
|                                                                                  | 5% DB       | MCDBR  | 18.8  | 19.1  | 19.9  | 20.1  | 20.0  | 19.5  | 18.3  | 18.5  | 19.4  | 21.3  | 22.4  | 18.9 |
|                                                                                  |             | MCWBR  | 13.0  | 12.0  | 10.6  | 10.1  | 8.2   | 7.1   | 6.8   | 6.7   | 7.9   | 10.1  | 13.5  | 13.3 |
|                                                                                  | 5% WB       | MCDBR  | 16.0  | 16.4  | 16.3  | 15.7  | 16.5  | 16.9  | 16.4  | 16.4  | 16.3  | 16.2  | 18.2  | 16.4 |
|                                                                                  |             | MCWBR  | 12.5  | 12.3  | 10.5  | 9.7   | 7.6   | 7.1   | 6.8   | 6.8   | 7.2   | 9.0   | 13.2  | 13.5 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.333  | 0.354 | 0.372 | 0.414 | 0.444 | 0.469 | 0.519 | 0.506 | 0.456 | 0.384 | 0.356 | 0.341 |      |
|                                                                                  | taud        | 2.536  | 2.484 | 2.428 | 2.313 | 2.265 | 2.236 | 2.125 | 2.160 | 2.265 | 2.454 | 2.475 | 2.529 |      |
|                                                                                  | Ebn at Noon | 286    | 289   | 290   | 279   | 270   | 262   | 248   | 251   | 260   | 274   | 274   | 277   |      |
|                                                                                  | Edh at Noon | 28     | 31    | 36    | 41    | 43    | 45    | 50    | 48    | 42    | 32    | 29    | 27    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 942    | 1113  | 1493  | 1826  | 2002  | 1945  | 1846  | 1750  | 1630  | 1426  | 1086  | 842   |      |
|                                                                                  | RadStd      | 82     | 155   | 128   | 150   | 114   | 155   | 136   | 140   | 122   | 158   | 97    | 76    |      |

Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | +1.28 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (27 neighbors) |  | +0.77 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +248  | +162  |

Nomenclature: See separate page